

1. Write expected output.

```
1 module tb;
2
3   bit [5:0] data;
4   covergroup cg;
5     coverpoint data {
6       bins a[] = {0, 2, 4, 6};
7       bins b[] = {[10:12], [15:16]};
8       bins c   = {[20:25]};
9       bins d[] = {30, [40:41], 50}; }
10
11   endgroup
12
13   cg cg_inst = new();
14
15   initial begin
16
17     int values[8] = '{2, 11, 15, 22, 30, 41, 18, 50};
18
19     foreach(values[i]) begin
20
21       #5 data = values[i];
22       cg_inst.sample();
23       $display("Sampled data = %0d", data);
24     end
25     $display("Coverage = %0.2f%%", cg_inst.get_coverage());
26   end
27 endmodule
28
```

2. Write the output.

```
29
30 module tb2;
31   bit [3:0] a, b;
32   bit en;
33   covergroup cg;
34     cp_a : coverpoint a {
35       bins a1[] = {1, 2};
36       bins a2   = {[5:6]} iff(en); }
37
38     cp_b : coverpoint b {
39       bins b1 = {3};
40       bins b2[] = {[7:8]}; }
41     cross cp_a, cp_b;
42
43   endgroup
44
45   cg c = new();
46
47   int avals[4] = '{1, 5, 6, 2};
48   int bvals[4] = '{3, 7, 8, 7};
49   bit evals[4] = '{0, 0, 1, 1};
50
51   initial begin
52     foreach(aval[s[i]) begin
53       a = avals[i];
54       b = bvals[i];
55       en = evals[i];
56       c.sample();
57       $display("a=%0d b=%0d en=%0b",a, b, en);
58     end
59
60     $display("Coverage = %0.2f%%",c.get_coverage());
61   end
62 endmodule
63
```

3. Write output.

```
1
2 module tb3;
3
4   bit clk, a, b;
5   always #5 clk = ~clk;
6
7   initial begin
8
9     clk = 0;
10    a = 1; b = 0;
11    #10 a = 0;
12    #10 b = 1;
13    #10 a = 1; b = 0;
14    #10 b = 1;
15    #20 $finish;
16  end
17
18  property p;
19
20    @(posedge clk)
21      a |-> ##[1:2] b;
22  endproperty
23
24  ap : assert property(p)
25      $display("PASS at %0t", $time);
26      else
27          $display("FAIL at %0t", $time);
28
29  endmodule
30
```

4. Write output.

```
32
33 module tb4;
34
35   bit clk, a, b, c;
36   always #5 clk = ~clk;
37
38   initial begin
39
40     clk = 0;  a = 1; b = 0; c = 0;
41
42     #10 a = 0;
43     #10 b = 1;
44     #10 b = 0;
45     #10 c = 1;
46     #10 a = 1; c = 0;
47     #10 a = 0;
48     #10 b = 1;
49     #10 c = 0;
50     #20 $finish;
51  end
52
53  sequence s;
54    b ##1 c;
55  endsequence
56
57  property p;
58    @(posedge clk)
59      a |-> ##1 s;
60  endproperty
61
62  ap : assert property(p)
63      $display("PASS at %0t", $time);
64      else
65          $display("FAIL at %0t", $time);
66  endmodule
67
```